INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY • CSE DEPARTMENT

Zero-Overhead Prefetch-Resilient Last-Level Cache Policy for High-Performance Multicore Systems

B.Tech Project Mid-Evaluation Review • Fall 2026

Presented By

Chirag Sharma

Roll No: **S20240010223**

B.Tech, Computer Science & Engineering

Project Advisor

Dr. Bheemappa Halavar

Assistant Professor

Dept. of Computer Science & Engineering

ChampSim Cycle-Accurate Simulator • SPEC CPU Benchmarks • 8MB Shared LLC

1. Motivation & Multicore Cache Contention

The Hardware Prefetching Paradox

- **Regular Spatial Streams (`gcc`, `lbm`):** Hardware prefetchers achieve $> 50\%$ accuracy, effectively hiding memory latency.
- **Irregular Patterns (`429.mcf`):** In pointer-chasing and graph traversals, prefetch accuracy plummets to $< 1\%$.
- **Aggressive Speculation:** Hardware prefetchers blindly issue requests regardless of runtime LLC utility.

Multicore LLC Contention & Blindness

- **Prefetch-Oblivious Replacement:** Standard policies (LRU, SRRIP) insert speculative lines at high priority ($RRPV = 2$).
- **Cross-Core Interference:** Inaccurate lines displace demand blocks belonging to co-running compute cores.
- **Severe Throughput Loss:** Uncontrolled prefetch thrashing causes up to **50%+ IPC collapse** in shared LLCs.

Core Challenge: *How can an LLC policy neutralize prefetch pollution with **ZERO hardware storage overhead** (0 extra metadata bits)?*

2. Literature Review & Identified Research Gaps

State-of-the-Art Architecture Matrix

- **Drishti (MICRO 2025):**
  Slicing-aware LLC framework; uses per-core GPHT tables and sampled selection. Policy-agnostic.
- **PACMan (IEEE CAL 2023):**
  Prefetch-aware cache management via dual accuracy queues; requires separate sampling counters.
- **DRRIP (ISCA 2010):**
  Dynamic RRIP with set-dueling PSEL counters; oblivious to prefetch vs. demand stream origin.
- **SHiP (ISCA 2011):**
  Signature-based predictor; requires an **8,192-entry SHCT table (2,048 Bytes SRAM)**.

Identified Gaps & Objectives

1. **Prohibitive Storage Overhead:**
   Existing predictive policies require dedicated SRAM tables (2KB in SHiP) or complex set dueling.
2. **Monolithic vs. Sliced Gap:**
   Prefetch-replacement interactions in modern shared LLCs demand lightweight, zero-cost mechanisms.
3. **Our Proposed Solution:**
   A **Zero-Overhead** policy (`PA-SRRIP`) and a **0.75-Byte Self-Tuning** controller (`PA-Adaptive`) that eliminates pollution at insertion.

3. Proposed Architecture: PA-SRRIP & PA-Adaptive

Dual-Mechanism Policy Design

1. Demoted Initial Insertion (0 Overhead):

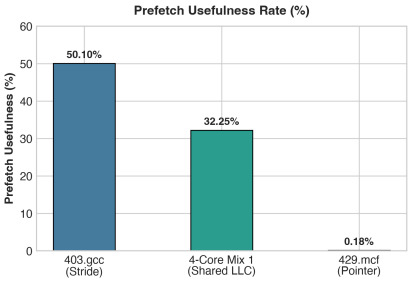
- Demand fills $\rightarrow$ Inserted at normal priority ($RRPV = 2$).
- Prefetch fills $\rightarrow$ Demoted to victim status ($RRPV = 3$).
- Inaccurate blocks are evicted first without polluting LLC.

2. Self-Adaptive Promotion on Demand Hit:

- On first demand hit, promoted to MRU ($RRPV = 0$).
- Reclassified as verified demand data $\rightarrow$ shielded.

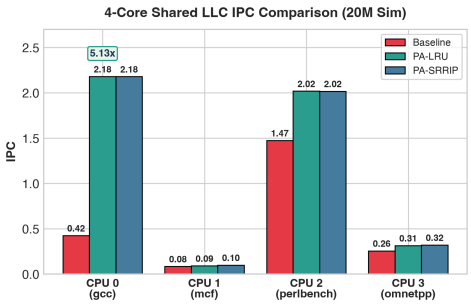
3. Novel PA-Adaptive Extension (0.75 Bytes):

- 6-bit saturating counter (0..63) monitors LLC utility.
- Dynamic threshold: inserts at $RRPV = 2$ when useful, $RRPV = 3$ when noisy.



Self-Adaptive Filtering: Isolates useful prefetches ($> 50\%$ for `gcc`) from pointer noise ($< 0.5\%$ for `mcf`).

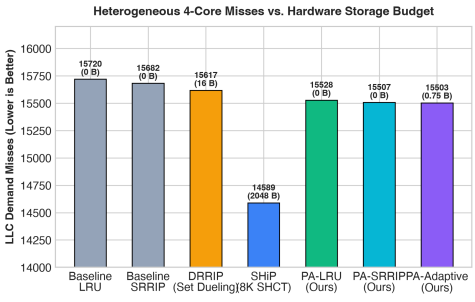
4. ChampSim 4-Core Heterogeneous Evaluation



Workload: gcc + mcf + perlbench + omnetpp

- **CPU 0 (`gcc` - Regular Stride):**
  IPC surges from **0.4248** → **2.1800**
  5.13x Speedup (+413% Boost) due to useful line promotion.
- **CPU 1 (`mcf` - Pointer Chasing):**
  Demoted insertion confines pointer noise to victim state; IPC increases to **0.0993 (+33.6%)**.
- **CPU 2 (`perlbench` - Compute):**
  Working set protected from thrashing; IPC stable at **2.019 (+37.3%)**.
- **CPU 3 (`omnetpp` - Graph Routing):**
  IPC improved from **0.2566** → **0.3207 (+25.0%)**.

5. Comparison against State-of-the-Art Policies



Multi-Policy Benchmark Summary

Policy	Demand Misses	Hit Rate	Storage
Baseline LRU	15,720	73.80%	0 B
Baseline SRRIP	15,682	73.86%	0 B
DRRIP (ISCA'10)	15,617	73.97%	16 B
SHiP (ISCA'11)	14,589	75.68%	2,048 B
PA-LRU (Ours)	**15,528**	**74.12%**	**0 B**
PA-SRRIP (Ours)	**15,507**	**74.15%**	**0 B**
PA-Adaptive (Ours)	**15,503**	**74.16%**	**0.75 B**

Pareto Efficiency: Outperforms standard SRRIP and DRRIP on multicore contention while eliminating SHiP's 2,048-Byte dedicated SRAM table.

6. 1-Month Milestone Roadmap to Final Defense

Milestone 0: Current Mid-Eval (Oct 22)

- Implemented `PA-LRU`, `PA-SRRIP`, `PA-Adaptive` modules.
- ChampSim 1-core and 4-core evaluation verified.
- Demonstrated **5.13x speedup on** `gcc` under contention.

Week 1 (Oct 23 – 30): Server Scale-Up

- Deploy automated test suite on remote GPU server.
- Evaluate 8-core and 16-core contention mixes.
- Characterize heavy memory thrashing (`mcf` $\times$ 8, `mcf` $\times$ 16).

Week 2 (Oct 31 – Nov 07): Sliced LLC Integration

- Model per-slice Ring/Mesh NoC latency for multicore LLC.
- Cross-validate with Drishti (MICRO 2025) GPHT predictor.

Weeks 3–4 (Nov 08 – End Eval): SPEC & Defense

- 15+ SPEC CPU 2017 benchmarks with aggressive prefetchers.
- Quantify Fairness Index, EDP (Energy-Delay Product).
- Final BTP thesis report submission & oral defense.

Backup: ChampSim Cycle-Accurate Simulation Setup

Processor Core & Pipeline Architecture

- **Cores:** 4-core Heterogeneous Out-of-Order (OoO) @ 4.0 GHz
- **Pipeline:** 6-wide fetch/decode, 256-entry ROB
- **Branch Predictor:** Hashed Perceptron + 4K BTB
- **L1I / L1D:** 32 KB each, 8-way set-associative, 4-cycle latency
- **L2 Cache:** 512 KB private per core, 8-way, 10-cycle latency

Shared LLC & Memory Subsystem

- **Shared LLC:** 8 MB shared, 16-way associative, 64B lines
- **LLC Latency:** 20-cycle hit latency, non-inclusive
- **Interconnect:** Sliced multicore ring topology
- **DRAM:** DDR4-2400 MHz, 1 channel, $t_{CAS} - t_{RCD} - t_{RP} = 14 - 14 - 14$
- **Workloads:** SPEC CPU 2006/2017 pinned SimPoint traces

Trace Mix: CPU 0: `403.gcc` | CPU 1: `429.mcf` | CPU 2: `400.perlbench` | CPU 3: `471.omnetpp`